ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

ICT 2205 Data Structures and Algorithms

Mid-semester test - Academic year 2025, semester 1

Time allowed: 1 hours Total marks: 50 Lecturer: Mrs J. Chibwe

Instructions: answer ALL questions. Write clearly. Show all working.

Question 1: Linked lists

(a) Inserting at the head of a singly linked list takes:

(b) Each node of a doubly linked list stores:

(c) Finding the k-th element of a linked list takes:

(d) A circular linked list's last node points to:

Question 2: Stacks and queues

(a) A stack follows which order?

(b) Which structure is used to evaluate postfix expressions?

(c) Breadth-first search uses a:

(d) Removing an element from a queue is called:

Question 3: Trees

(a) In-order traversal of a binary search tree visits keys in:

(b) The height of a balanced binary tree with n nodes is about:

(c) A binary heap is usually stored in:

(d) Which tree keeps itself balanced using rotations based on height differences?

END OF PAPER - generated sample for the ZutLib prototype